BEST FIT

```
[rene@localhost ~]$ cd memory
[rene@localhost memory]$ vi bestFit.c
[rene@localhost memory]$
```

```c
#include<stdio.h>
int main(){
        int m,n;
        printf("Enter number of memory blocks:\n");
        scanf("%d",&m);
        int blockSize[m],remaining[m];
        printf("Enter size of each block:\n");
        for(int i=0;i<m;i++){
                scanf("%d",&blockSize[i]);
                remaining[i] = blockSize[i];
        }
        printf("ENter number of processes:\n");
        scanf("%d",&n);
        int processSize[n];
        printf("Enter size of each process:\n");
        for(int i=0;i<n;i++){
                scanf("%d",&processSize[i]);
        }
        for(int i=0;i<n;i++){
                int bestIndex = -1;
                for(int j=0;j<m;j++){
                        if(remaining[j] >= processSize[i]){
                                if(bestIndex==-1 || remaining[j] < remaining[bestIndex]){
                                        bestIndex = j;
                                }
                        }
                }
                if(bestIndex != -1){
                        int fragment = remaining[bestIndex] - processSize[i];
                        printf("Process %d of size %d is allocated to Block %d of size %d with fragment %d\n",i+1,processSize[i],bestIndex+1,blockSize[bestInd
ex], fragment);
                        remaining[bestIndex] -= processSize[i];
                }else{
                        printf("Process %d of size %d is not allocated\n", i+1,processSize[i]);
                }
        }
        return 0;
}
~
~
~
~
~
~
                                                                                                                                38,0-1        All
```

```
[rene@localhost ~]$ cd memory
[rene@localhost memory]$ vi bestFit.c
[rene@localhost memory]$ gcc bestFit.c
[rene@localhost memory]$ ./a.out
Enter number of memory blocks:
5
Enter size of each block:
100
500
200
300
600
ENter number of processes:
4
Enter size of each process:
212
417
112
426
Process 1 of size 212 is allocated to Block 4 of size 300 with fragment 88
Process 2 of size 417 is allocated to Block 2 of size 500 with fragment 83
Process 3 of size 112 is allocated to Block 3 of size 200 with fragment 88
Process 4 of size 426 is allocated to Block 5 of size 600 with fragment 174
[rene@localhost memory]$
```